

2(a)

$$(i) S \rightarrow +SS \mid *SS \mid S \mid a$$

string: $+a*aa$

$$S \rightarrow +SS$$

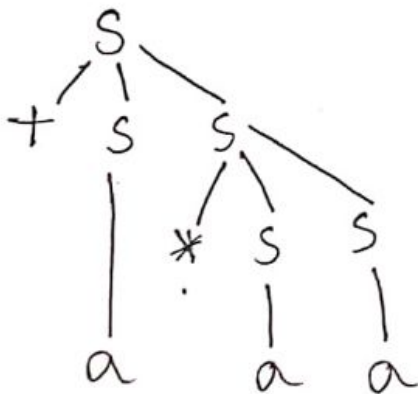
$$\rightarrow +S*SS$$

$$\rightarrow +a*SS$$

$$\rightarrow +a*aS$$

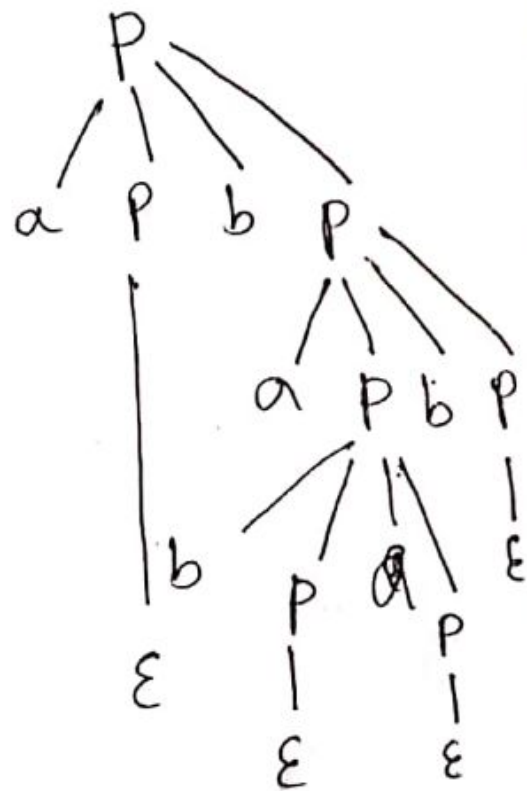
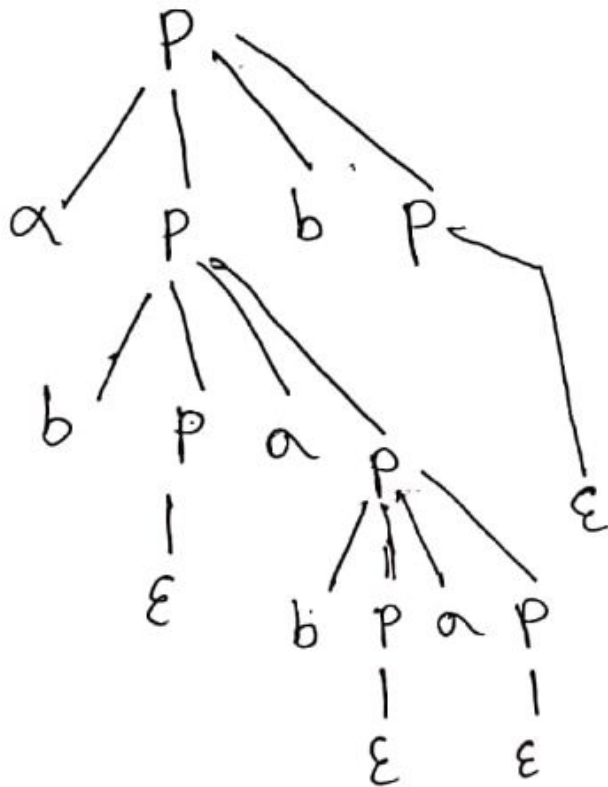
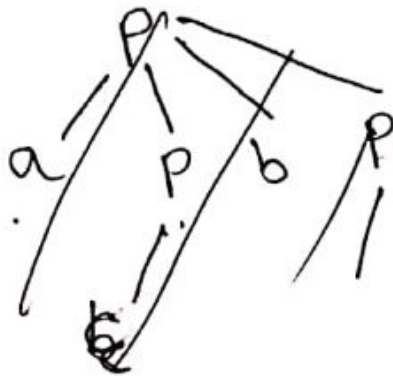
$$\rightarrow +a*as$$

(ii)



2(b)

String: ab abab

$$P \rightarrow aPbP|bPaP|\epsilon$$


2(c)

(i) ~~{ba, bb, aab}~~

(ii)

(iii)

(i) Regular expression: b^*a^+

Accepted strings = {ba, bbaa}

(ii) Regular expression: .

Accepted: {aaaa, babbab}

(iii) Regular expression: $(aa)^*b$

Accepted strings = {b, aab}

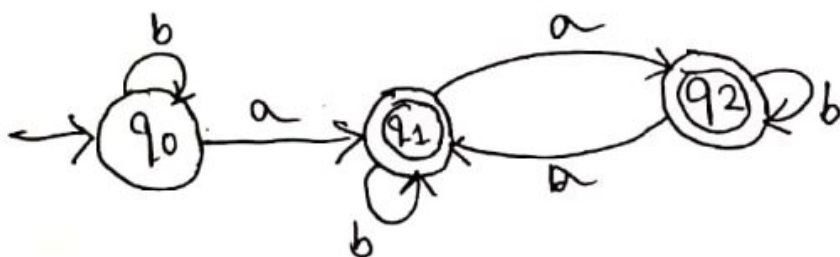
(iv) Regular expression $(011)(011)^*$

Accepted strings = {00, 11}

3(a)

~~Step 1~~

Remove all the states, those are not reachable from initial state.



~~Step 2~~

	a	b
→ q ₀	*q ₁	q ₀
* q ₁	*q ₂	*q ₁
* q ₂	*q ₁	*q ₂

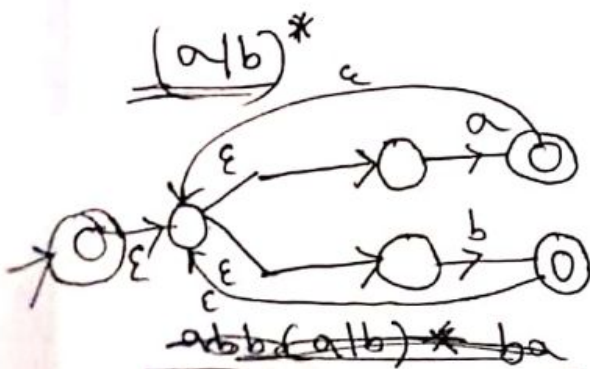
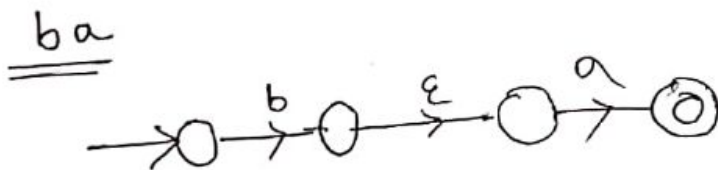
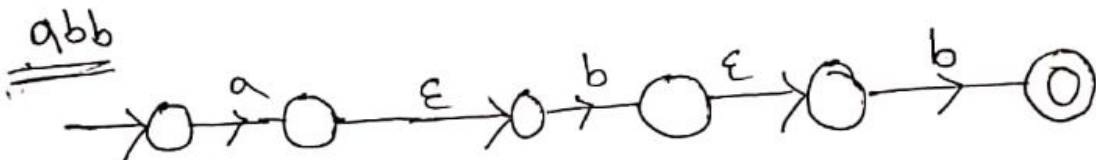
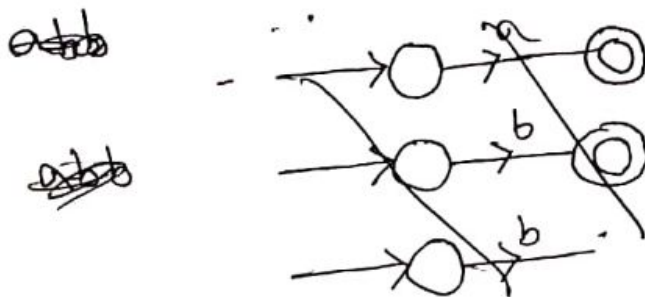
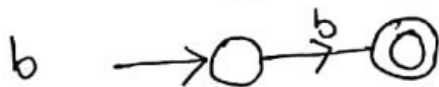
0 equivalent state = [q₁, q₂] [q₀]

1 equivalent state = [q₁ q₂] [q₀]

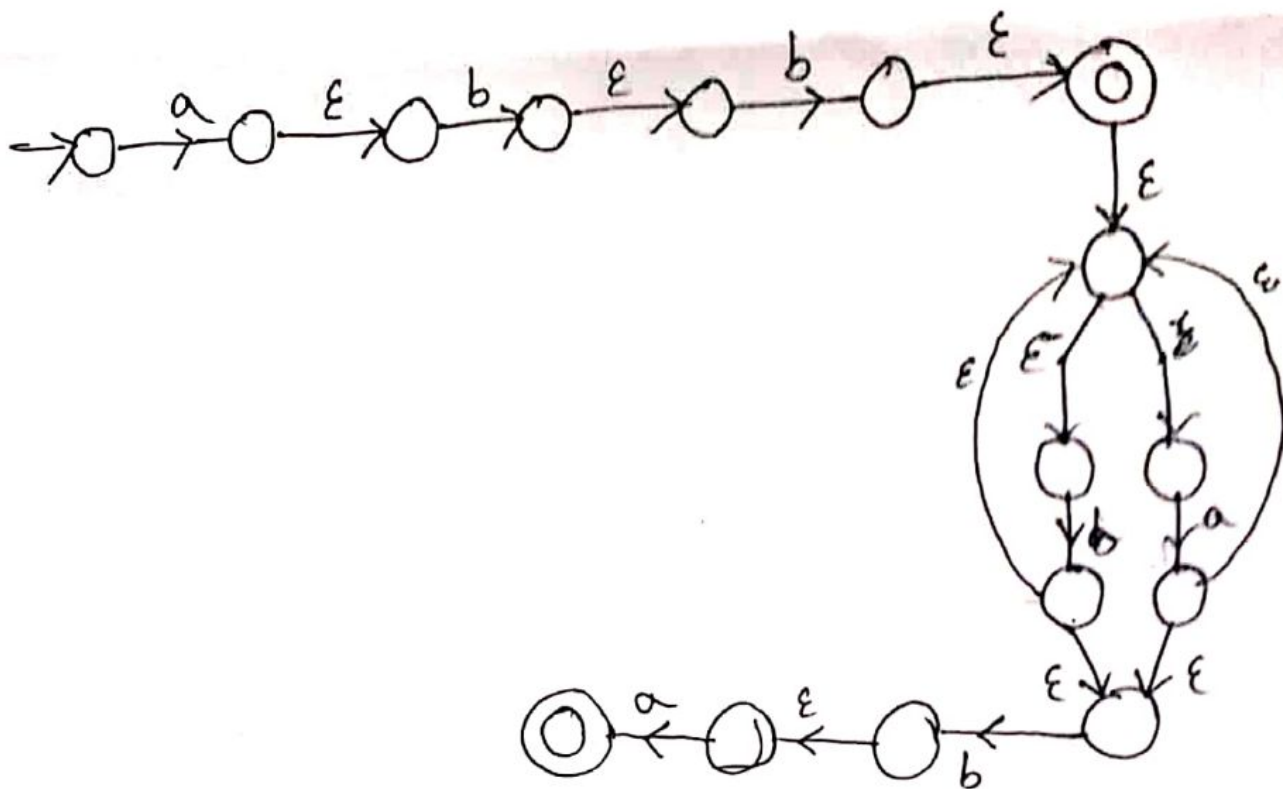


3(b)

$abb(a|b)^*ba$



$a \rightarrow b \rightarrow a \rightarrow b$

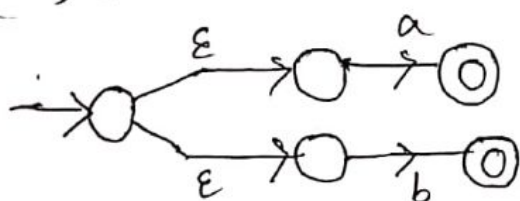


b or

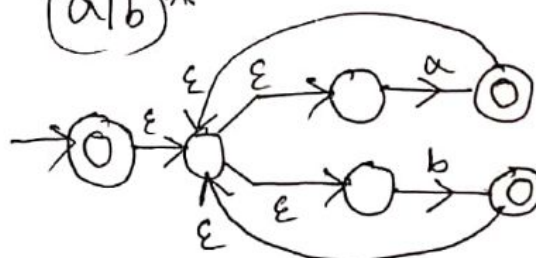
$$(a|b)^* a(a|b)(a|b)$$



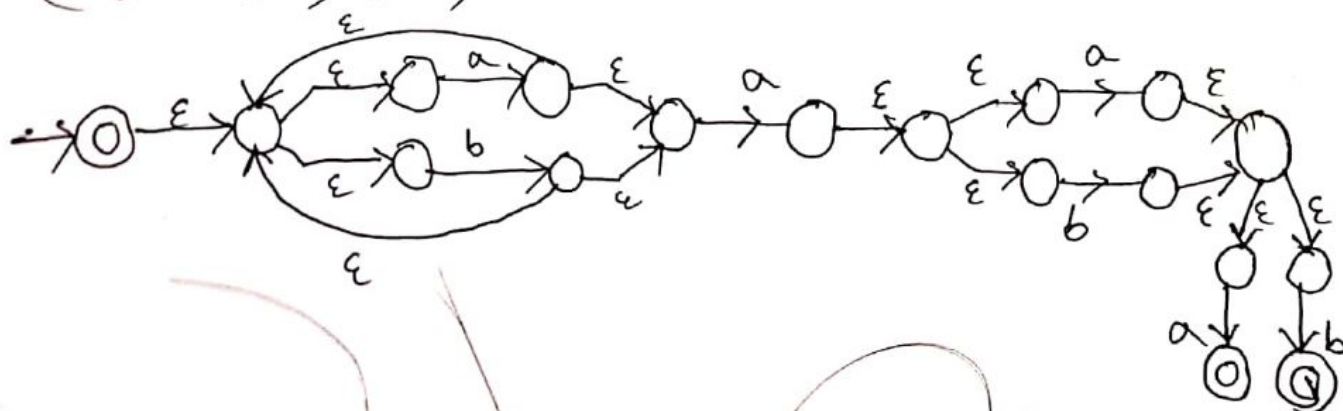
$(a|b)$:



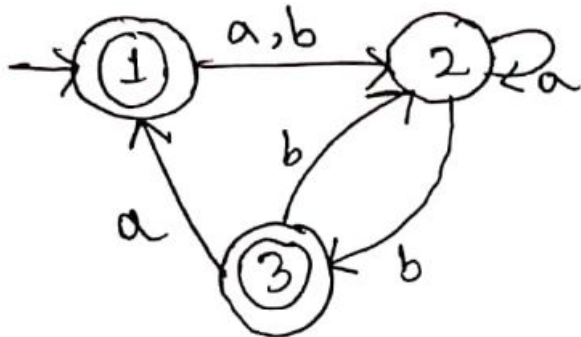
$(a|b)^*$



$$(a|b)^* a(a|b)(a|b)$$



3(c)



S.T.T

		a	b
→ *	1	{2}	{2}
	2	{2}	{3}
*	3	{1}	{2}

DFA S.T.T

	a	b
→ * [1]	[2]	[2]
• [2]	[2]	[3]
* [3]	[1]	[2]

